Time series: estimation

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The homework is provided on the last slide

Estimation of characteristics

For a stationary process x_t in discrete time, using a *single realization* of this process, we can estimate

- the mean $\mu = \mathbf{E}x_t$,
- the covariance function

$$R(k) = \mathbf{E}(x_t - \mu)(x_{t+k} - \mu), \ k = 0, \pm 1, \pm 2, \ldots$$

- the spectral density

$$f(\lambda) = \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} R(k) \cos(\lambda k), \ \lambda \in [-\pi, \pi].$$

Estimation of the mean, unbiasedness

Let $x_1, x_2, x_3, \ldots, x_n$ be a realization of a stationary process with mean μ and the covariance function $R(k)$.

Theorem. The estimator

$$\bar{X}_n = \frac{1}{n}\sum_{j=1}^{n} x_j$$

is an unbiased estimator of the mean μ.

Proof. Since $\mathbf{E}x_j = \mu$ for all j, we have

$$\mathbf{E}\bar{X}_n = \mathbf{E}\Big(\frac{1}{n}\sum_{j=1}^{n} x_j\Big) = \frac{1}{n}\sum_{j=1}^{n} \mathbf{E}x_j = \frac{1}{n}\sum_{j=1}^{n} \mu = \mu.$$

Estimation of the mean, variance

Theorem. If $f(0) \neq 0$ and $f(0) < \infty$, then

$$n\mathrm{Var}(\bar{X}_n) \to 2\pi f(0) \text{ as } n \to \infty.$$

Proof. Since

$$\mathrm{Var}(\bar{X}_n) = \mathrm{Cov}(\bar{X}_n, \bar{X}_n) = \frac{1}{n^2}\mathrm{Cov}\left(\sum_i x_j, \sum_j x_j\right)$$

$$= \frac{1}{n^2}\sum_{i=1}^{n}\sum_{j=1}^{n}\mathrm{Cov}(x_i, x_j) = \frac{1}{n^2}\sum_{i=1}^{n}\sum_{j=1}^{n}R(j-i)$$

$$= \frac{1}{n^2}\sum_{k=-(n-1)}^{n-1}(n-|k|)R(k) = \frac{1}{n}\sum_{k=-(n-1)}^{n-1}\left(1-\frac{|k|}{n}\right)R(k),$$

we have

$$\lim_{n\to\infty} n\mathrm{Var}(\bar{X}_n) = \lim_{n\to\infty}\sum_{k=-(n-1)}^{n-1}\left(1-\frac{|k|}{n}\right)R(k)$$

$$= \sum_{k=-\infty}^{\infty}\lim_{n\to\infty}\left(1-\frac{|k|}{n}\right)R(k) = \sum_{k=-\infty}^{\infty}R(k) = 2\pi f(0).$$

Estimation of the mean, normality

Theorem. If $\mathbf{E}|x_t^3| < \infty$, $f(0) \neq 0$ and the mixing coefficient

$$\alpha(\tau) = \sup_{A \in \mathcal{F}\{x_s\}_{s\in(-\infty,t)}, B \in \mathcal{F}\{x_s\}_{s\in[t+\tau,\infty)}} \left|P(A\cap B) - P(A)P(B)\right|$$

satisfies $\alpha(\tau) = O(1/\tau^{1+\varepsilon})$ as $\tau \to \infty$, $\varepsilon > 2/3$, then

$$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{D} N(0, 2\pi f(0))$$

as $n \to \infty$.

Theorem. The asymptotic confidence interval for the mean μ is

$$\left(\bar{X}_n - z_{1-\alpha/2}\sqrt{\frac{2\pi f(0)}{n}}, \bar{X}_n + z_{1-\alpha/2}\sqrt{\frac{2\pi f(0)}{n}}\right).$$

Estimation of covariance function $R(k)$

To estimate $R(k)$, we can use the following two estimators:

$$\hat{R}(k) = \frac{1}{n}\sum_{j=1}^{n-k}(x_j - \bar{X}_n)(x_{k+j} - \bar{X}_n)$$

and

$$\tilde{R}(k) = \frac{1}{n-k}\sum_{j=1}^{n-k}(x_j - \bar{X}_n)(x_{k+j} - \bar{X}_n).$$

These estimators are asymptotically unbiased.

The estimator $\tilde{R}(k)$ has smaller bias and larger MSE than $\hat{R}(k)$.

The sequence $\hat{\rho}(k) = \hat{R}(k)/\hat{R}(0)$, $k = 0, 1, 2, \ldots$, is called a **correlogram** and provides an estimator of the correlation function $\rho(k) = R(k)/R(0)$.

The difference k is called the **lag**.

Example of computing $\hat{R}(k)$

Task. Estimate the mean and the covariance function for the time series 4, 1, 4, 5, 1.

Solution: $n = 5$, $\bar{X}_n = (4+1+4+5+1)/5 = 3$,

t	$x_t - \bar{X}_n$	$x_{t+1} - \bar{X}_n$	$x_{t+2} - \bar{X}_n$	$x_{t+3} - \bar{X}_n$	$x_{t+4} - \bar{X}_n$
1	1	-2	1	2	-2
2	-2	1	2	-2	
3	1	2	-2		
4	2	-2			
5	-2				

$\hat{R}(0) = (1^2 + (-2)^2 + 1^2 + 2^2 + (-2)^2)/5 = 14/5$

$\hat{R}(1) = (1 \cdot (-2) + (-2) \cdot 1 + 1 \cdot 2 + 2 \cdot (-2))/5 = -6/5$

$\hat{R}(2) = (1 \cdot 1 + (-2) \cdot 2 + 1 \cdot (-2))/5 = -5/5 = -1$

$\hat{R}(3) = (1 \cdot 2 + (-2) \cdot (-2))/5 = 6/5$

$\hat{R}(4) = (1 \cdot (-2))/5 = -2/5$

$\hat{R}(s) = 0$ for $s = 5, 6, \ldots$

R code for correlogram

Task. Depict the correlogram for the time series 4,1,4,5,1.

® *x=c(4,1,4,5,1); n=length(x); mu=sum(x)/n; y=x-mu;*
covg=rep(0,n); corrg=rep(0,n);
for(i in (1:n))
{
*covg[i]=sum(y[1:(n+1-i)]*y[i:n])/n;*
corrg[i]=covg[i]/covg[1];
}
print(rbind(c(0:(n-1)),covg,corrg))
barplot(corrg,names.arg=c(0:(n-1)), main="correlogram",
ylim=c(-1,1))

Here *covg[i]* contains $R(i-1)$, $i = 1, 2, \ldots$.

Task. Depict the correlogram for a realization of the white noise process with mean 0 and variance 3^2, of length 150.

® *x=rnorm(150,0,3); And proceeding as described above.*

Correlogram for MA process

Simulate the MA process $x_k = 0.9\epsilon_k - 0.6\epsilon_{k-1} + 0.3\epsilon_{k-2} + 0.1\epsilon_{k-3}$, where ϵ_k is white noise with mean 0 and variance 0.8^2, $k = 4, 5, \ldots, 90$. Depict the correlogram of time series x_k.

```
® eps=rnorm(90,0,0.8);
x=0.9*eps[4:90]-0.6*eps[3:89]+0.3*eps[2:88] +0.1*eps[1:87];
n=length(x);  mu=sum(x)/n;  y=x-mu;
covg=rep(0,n); corrg=rep(0,n);
for(i in (1:n))
{
  covg[i]=sum(y[1:(n+1-i)]*y[i:n])/n;
  corrg[i]=covg[i]/covg[1];
}
par(mfrow=c(2,1));
plot(c(1:n),x,main="time series: MA process",pch=19);
barplot(corrg,names.arg=c(0:(n-1)), main="correlogram",
    ylim=c(-1,1))
```

Estimation of the spectral density

Recall that a stationary discrete process can be considered as a linear combination of sinusoids with different frequencies.

The larger value of the spectral density $f(\lambda)$ means the larger contribution of the frequency λ.

Consider a time series such that $\frac{1}{n}\sum_{j=1}^{n} x_j = 0$.

The function $P(\lambda) = T_c^2(\lambda) + T_s^2(\lambda)$, where

$$T_c(\lambda) = \frac{1}{\sqrt{n}} \sum_{j=1}^{n} x_j \cos(\lambda j)$$

$$T_s(\lambda) = \frac{1}{\sqrt{n}} \sum_{j=1}^{n} x_j \sin(\lambda j)$$

is called a **periodogram** and serves an estimator of $2\pi f(\lambda)$.

Periodogram

Define $\lambda_j = 2\pi j/n,\ j = 1, \ldots, n.$

λ is called an **angular frequency**.
$\nu = \frac{\lambda}{2\pi}$ is called an **ordinary frequency**.

The periodogram $P(\lambda)$, $\lambda \in [0, \pi]$ is depicted by drawing $P(\lambda_1), \ldots, P(\lambda_n)$.

Theorem. The periodogram $P(\lambda)$ can be rewritten as $P(\lambda) = \frac{1}{n}\left|\sum_{j=1}^{n} x_j e^{i\lambda j}\right|^2$, where i is the imaginary unit, and

$$P(\lambda) = \sum_{k=-(n-1)}^{n-1} \hat{R}(k) e^{i\lambda k} = \sum_{k=-(n-1)}^{n-1} \hat{R}(k) \cos(\lambda k).$$

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R-code for periodogram

Task. Compute and depict the periodogram for a realization of white noise with mean 0 and variance 1.5^2, of length 200.

```
® x=rnorm(200,0,1.5);
n=length(x); y=x-mean(x);
nu=c(0:(n/2-1))/n;
lambda=2*pi*nu;
Tc=rep(0,length(nu)); Ts=rep(0,length(nu));
for( j in (1:length(nu)))
{
Tc[j]=sum(y*cos((1:n)*lambda[j]))/sqrt(n);
Ts[j]=sum(y*sin((1:n)*lambda[j]))/sqrt(n);
}
I=Tc*Tc+Ts*Ts;
P=(4/n)*I;
plot(nu, P, type="l", xlab="Frequency",
    ylab="Scaled Periodogram")
```

R-code for periodogram via FFT

Since the temporal and spectral representations of a time series are connected by the Fourier transform, the function of Fast Fourier Transform (FFT) from the R package can be used.

```
® drawPlots=function(xts,title="time series")
{
n=length(xts);  y=xts-mean(xts);
I = abs(fft(y))**2/n;
P = (4/n)*I[1:(n/2)];
nu = c(0:(n/2-1))/n;
par(mfrow=c(2,1));    plot(c(1:n), xts, main=title,pch=20);
plot(nu, P, type="l", xlab="Frequency",
     ylab="Scaled Periodogram"); return(list(nu,P));
}
x=rnorm(200,0,1.5);
pdata=drawPlots(x,"time series: white noise")
```

Examples

Compute the periodogram for time series

- $x_k = \sin(2\pi k/50) + \epsilon_k,\ k = 1, \ldots, 200,$
- $x_k = \sin(2\pi k/50) + 2.5\epsilon_k,\ k = 1, \ldots, 200,$
- $x_k = \sin(2\pi k/50) + 3.8\epsilon_k,\ k = 1, \ldots, 200,$
- $x_k = \sin(2\pi k/22) + \epsilon_k,\ k = 1, \ldots, 200,$
- $x_k = \sin(2\pi k/50) + \sin(2\pi k/22) + \epsilon_k,\ k = 1, \ldots, 200,$
- $x_k = \sin(2\pi k/50) + 3\sin(2\pi k/22) + \epsilon_k,\ k = 1, \ldots, 200,$
- $x_k = \sin(2\pi k/50) + 0.3\sin(2\pi k/22) + \epsilon_k,\ k = 1, \ldots, 200,$
- $x_k = \sin(2\pi k/83) + \sin(2\pi k/50) + \sin(2\pi k/22) + \epsilon_k,$
- $x_k = [1.5 + \sin(2\pi k/83)]\sin(2\pi k/22) + \epsilon_k,$

where ϵ_k is white noise with mean 0 and variance 1.

Examples, 2

```
® n=200; tm=c(1:n);
x=sin(2*pi*tm/50)+2.5*rnorm(n,0,1);
p1=drawPlots(x,"time series: sin(2pik/50)+2.5WN")
x=sin(2*pi*tm/50)+3.8*rnorm(n,0,1);
p2=drawPlots(x,"time series: sin50+3.8WN")
x=sin(2*pi*tm/22)+rnorm(n,0,1);
p3=drawPlots(x,"time series: sin22+WN")
x=sin(2*pi*tm/50)+sin(2*pi*tm/22)+rnorm(n,0,1);
p4=drawPlots(x,"time series: sin50+sin22+WN")
x=sin(2*pi*tm/50)+3*sin(2*pi*tm/22)+rnorm(n,0,1);
p5=drawPlots(x,"time series: sin50+3sin22+WN")
x=sin(2*pi*tm/50)+0.3*sin(2*pi*tm/22)+rnorm(n,0,1);
p6=drawPlots(x,"time series: sin50+0.3sin22+WN")
x=(1.5+sin(2*pi*tm/83))*sin(2*pi*tm/22)+rnorm(n,0,1);
p7=drawPlots(x,"time series: (1.5+sin83)sin22+WN")
x=(1.5+sin(2*pi*tm/83))*sin(2*pi*tm/22)+0.2*rnorm(n,0,1);
p8=drawPlots(x,"time series: (1.5+sin83)sin22+.2WN")
```

Homework

Use AI tools to explore the practical and theoretical aspects of time series estimation.

- **Conceptual Exploration:** Ask an AI to explain why the sample mean $\bar{X}_n$ is an unbiased estimator for a stationary process, and how its variance relates to the spectral density at zero frequency.
- **Estimator Comparison:** Use AI to generate R code that compares the "biased" $(1/n)$ and "unbiased" $(1/(n-k))$ estimators for the autocovariance function. Ask the AI to simulate a process and plot both to see which is more stable for large lags.
- Solve exam-like exercises for L7.